AWS Cloud Foundations

Compute • Storage • Identity

COMPUTE STORAGE IDENTITY

AWS Compute Essentials

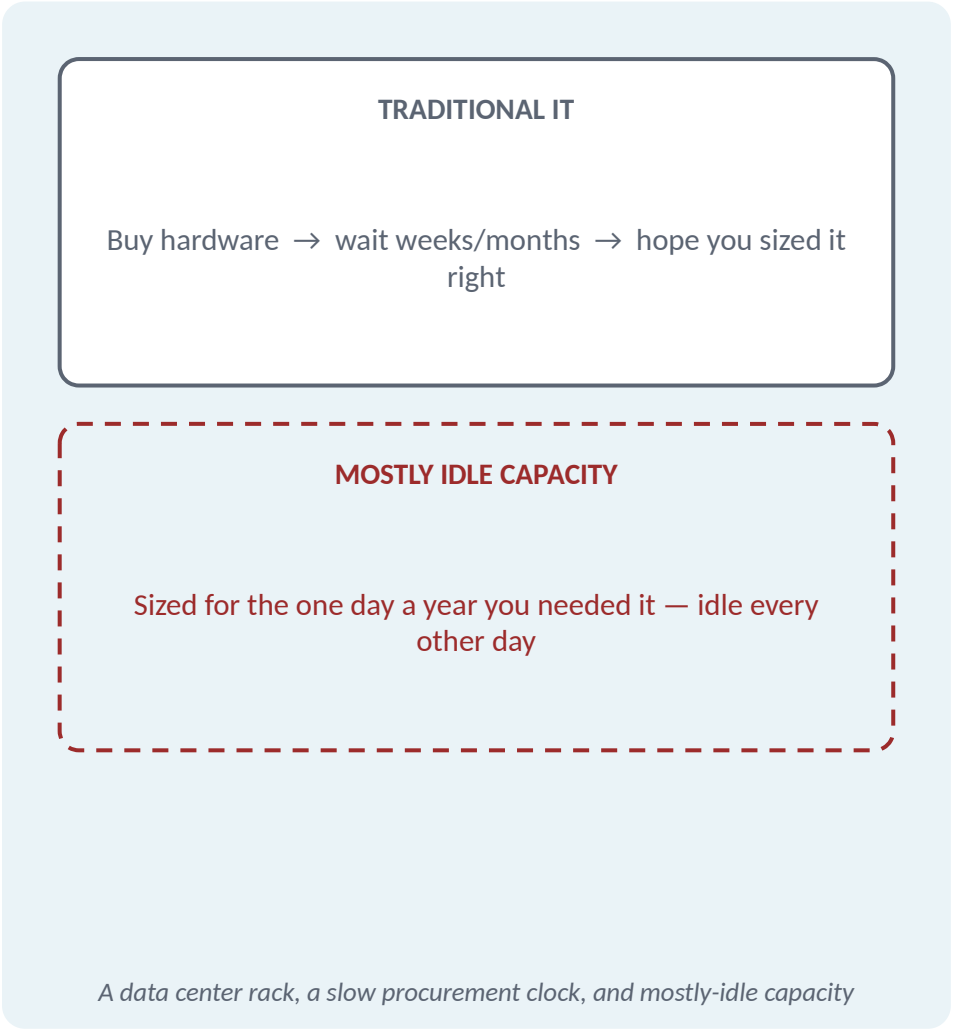
WHAT'S AHEAD

- Why the cloud, and the AWS Shared Responsibility Model
- AWS's service landscape — where EC2, S3, and IAM fit
- AWS global infrastructure: Regions & Availability Zones
- Compute & networking essentials: EC2, VPC, Security Groups
- Basic Auto Scaling groups

PROBLEM

Why Not Just Buy Hardware?

- Traditional infrastructure asks a team to commit to hardware before it knows what it actually needs
- Real procurement and setup takes real time — a project stalls waiting for servers to arrive and be configured
- To survive an unpredictable traffic spike, a team over-provisions for its worst day
- Most of that capacity sits idle the rest of the time; reaching a new geography means building infrastructure there too



A data center rack, a slow procurement clock, and mostly-idle capacity

SOLUTION

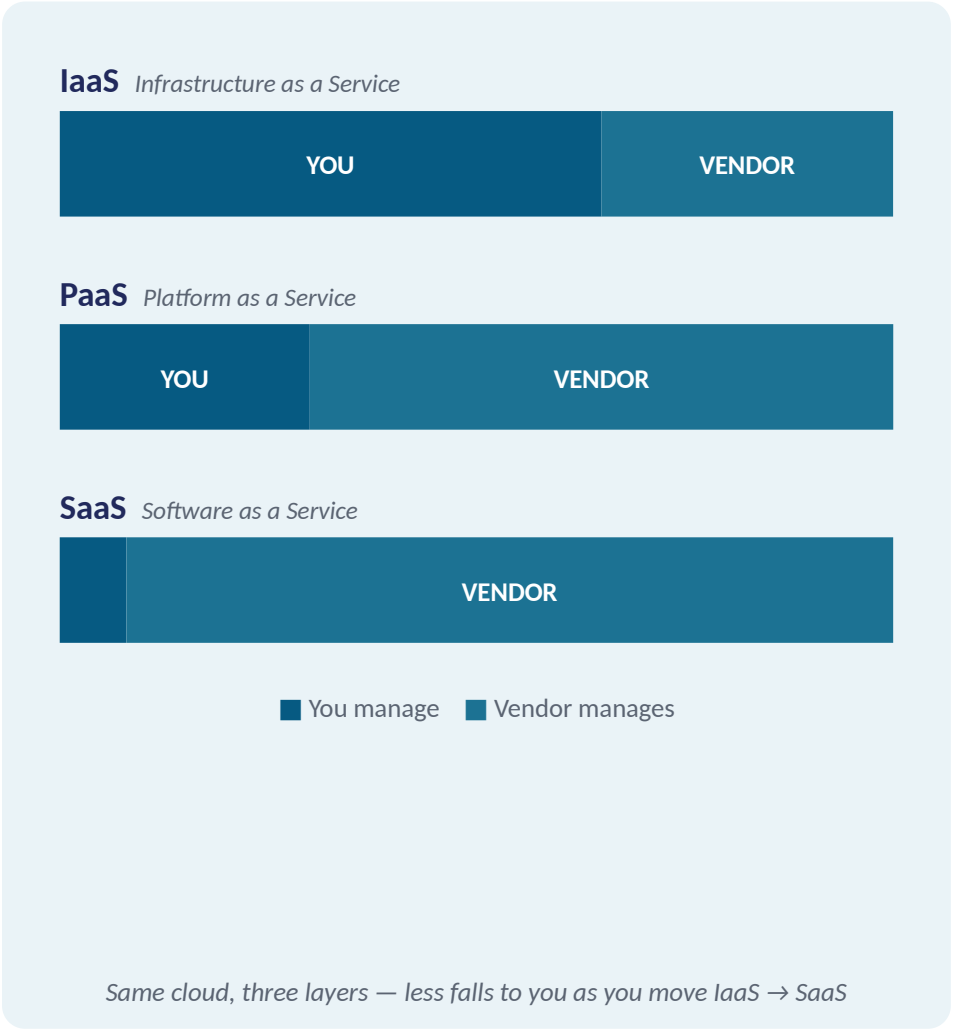
Pay-as-You-Go Cloud Infrastructure

- Cloud model: pay-as-you-go, access compute, storage, and networking on an as-needed basis
- Provision in minutes instead of months
- Scale up or down instantly as real demand changes
- Reach global users without ever building a physical data center

A dial that scales instantly, and reach without new data centers

Cloud Service Models: IaaS, PaaS, SaaS

- "The cloud" isn't one offering — IaaS, PaaS, and SaaS each draw the you-manage / vendor-manages line at a different layer
- IaaS (Infrastructure as a Service): rent raw compute, storage & networking — you still manage the OS, runtime & app (AWS EC2, S3 — today's focus)
- PaaS (Platform as a Service): vendor also manages the OS & runtime — you just deploy code (AWS Elastic Beanstalk, Heroku)
- SaaS (Software as a Service): vendor runs the entire application — you just use it (Gmail, Salesforce)

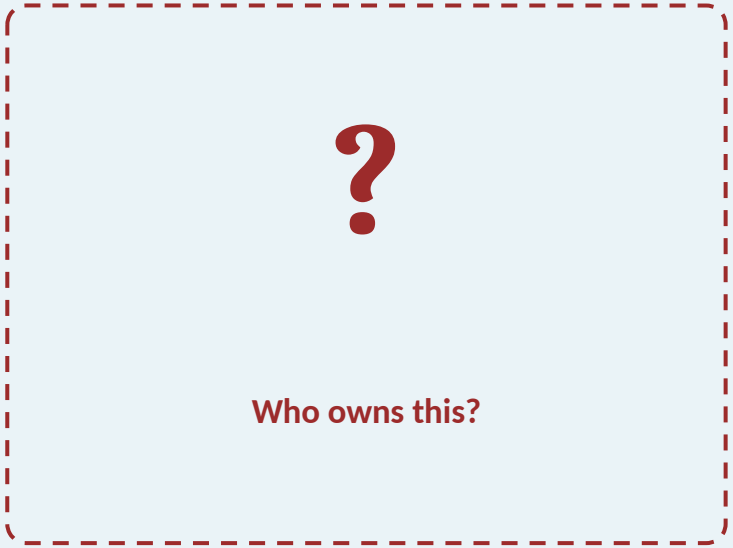


Same cloud, three layers — less falls to you as you move IaaS → SaaS

PROBLEM

Who's Responsible for Security Now?

- Moving infrastructure to AWS means AWS now runs the physical hardware, not your team
- Security doesn't just disappear as a concern because someone else owns the servers
- A team needs to know exactly what it's still on the hook for, and what it isn't
- Without a clear line, a real gap (an open port, a public bucket) gets missed until it's exploited

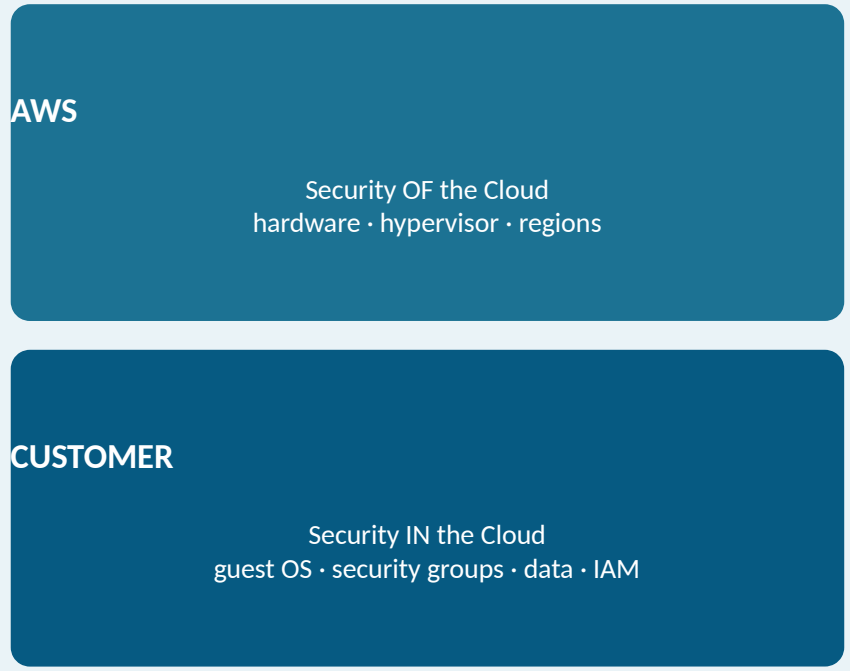


A question mark over AWS's infrastructure — who secures what?

SOLUTION

The AWS Shared Responsibility Model

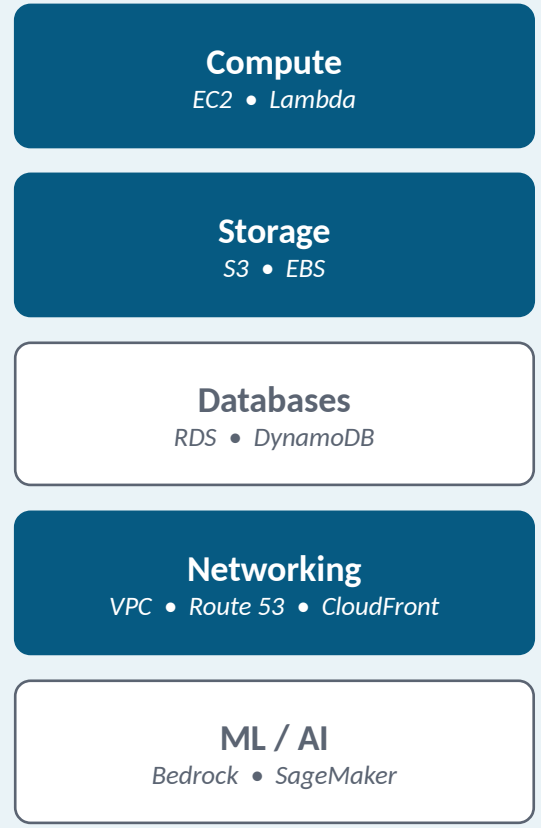
- AWS secures the cloud: physical facilities, hardware, networking, hypervisor, host OS, Regions/AZs
- Customer secures what's in the cloud: guest OS & patches, security group rules, data & encryption, IAM
- The split shifts by service — EC2 leaves more to the customer than a managed service like S3
- Today's defaults (security groups deny inbound, S3 blocks public access) are the customer's half of this line



AWS's half vs. the customer's half of the security line

AWS at a Glance

- AWS groups services into categories: Compute, Storage, Databases, Networking & Content Delivery, Machine Learning/AI
- Security & Identity (where IAM lives) cuts across every category
- Today's foundation: EC2 (Compute), VPC (Networking), S3 (Storage), IAM (Security & Identity)



AWS service categories, with EC2/VPC/S3/IAM highlighted

The Wider AWS Landscape

Where EC2, VPC, S3, and IAM sit among AWS's most common services, by category

Compute
EC2, Lambda, ECS

Storage
S3, EBS, EFS

Databases
RDS, DynamoDB, Aurora

Networking & Content Delivery
VPC, Route 53, CloudFront

Security, Identity & Compliance
IAM, KMS, WAF

Machine Learning / AI
Bedrock, SageMaker, Rekognition

Analytics
Athena, Kinesis, Glue

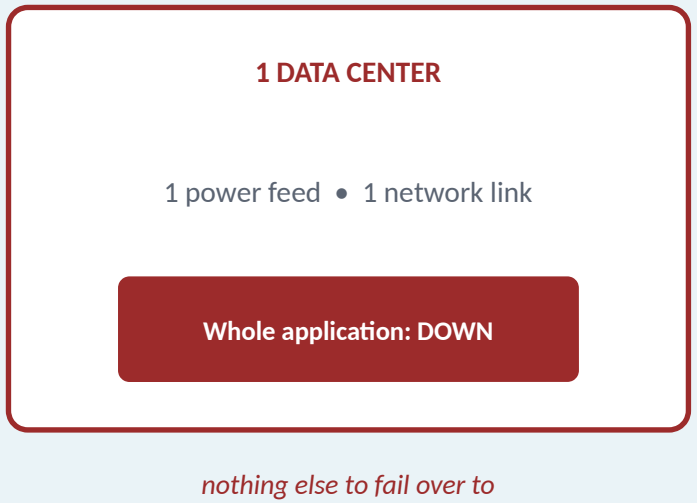
Management & Governance
CloudWatch, CloudTrail, Systems Manager

Highlighted: where this week's work lives (Compute/EC2, Networking/VPC, Storage/S3, Security/IAM)

Why Put All Your Eggs in One Data Center?

PROBLEM

- An application has to run somewhere physically real: one building, one power feed, one network link
- If that single "somewhere" fails, a power outage, a fire, a cut fiber line, the whole application goes down
- No amount of good code protects against losing the one physical place it all runs
- Before touching the console, the team needs a mental model for where anything they build actually runs

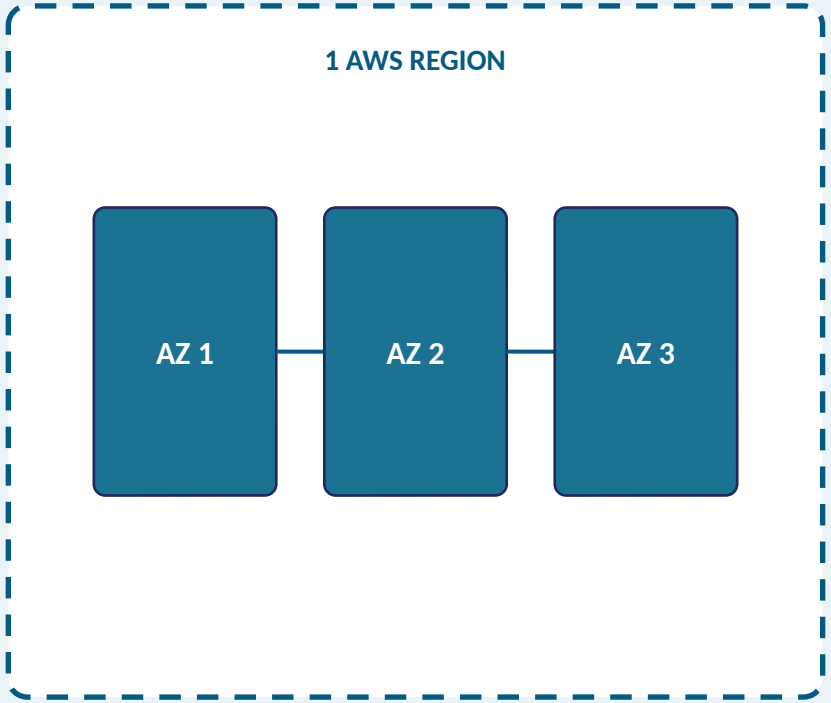


One data center, one bad day, the whole application down

SOLUTION

AWS Regions & Availability Zones

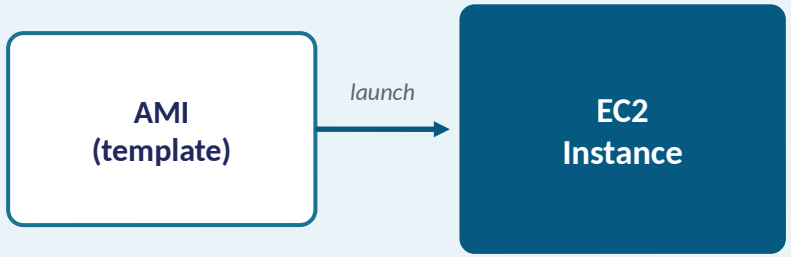
- Region: a physical location where AWS clusters data centers (39 regions today)
- Every region has 3+ Availability Zones (AZs) — 124 AZs total
- An AZ = one or more data centers with its own power, networking, connectivity
- Far enough apart to survive an outage, close enough for low-latency links
- Your team's Region choice today is sticky — pick one and keep it all week



One region containing three Availability Zones

EC2 Instances & AMIs

- EC2 = Amazon's virtual server service
- Instance = one running virtual machine
- AMI (Amazon Machine Image) = the template an instance launches from
- Instance type sets vCPU, memory, network, and cost — resizable later
- Key pair = the credential used to connect over SSH



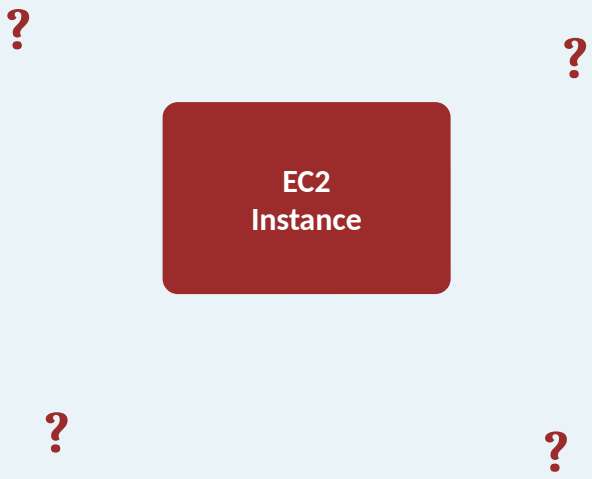
Launching an EC2 instance from an AMI template

HANDS-ON Launch an EC2 instance from an AMI

PROBLEM

Whose Network Is My Instance On?

- Every EC2 instance has to live somewhere on a network the moment it launches
- A wide-open, shared, or unmanaged network means anything could potentially reach it
- Before a security group can filter traffic, something has to define the network itself
- That question — which network, which IP range, which zone — has to be answered first

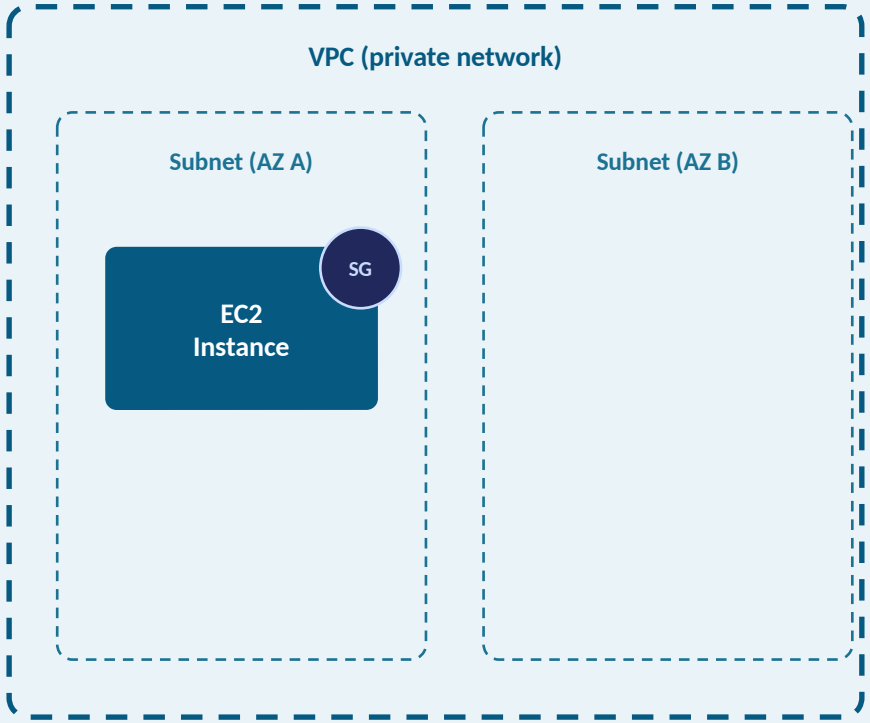


A lone EC2 instance with no defined network boundary around it

SOLUTION

Amazon VPC: Your Private Network

- A VPC is your own logically isolated private network inside AWS
- Subnets subdivide a VPC; each subnet lives in exactly one Availability Zone
- Every account gets a default VPC per Region, pre-configured so instances launch immediately
- Security groups (next) filter traffic on top of whichever VPC and subnet an instance is in

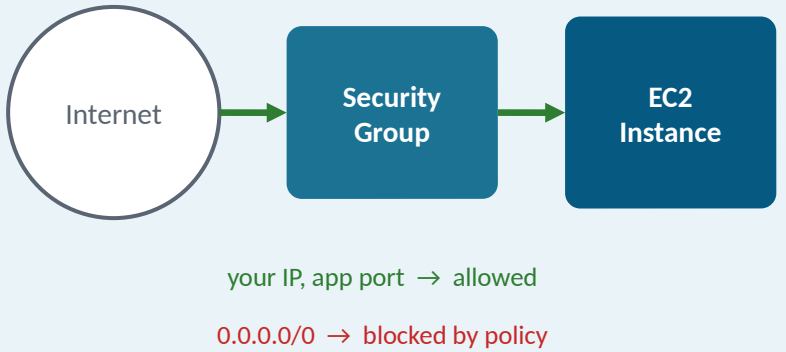


A VPC containing subnets across two AZs, with an EC2 instance and its security group

HANDS-ON Part of: Launch an EC2 instance from an AMI

Security Groups: Your Instance's Firewall

- A security group is a stateful virtual firewall for your instance
- Default: no inbound traffic allowed, all outbound traffic allowed
- An inbound rule = Type + Port range + Source
- AWS guidance: never open SSH/RDP to 0.0.0.0/0



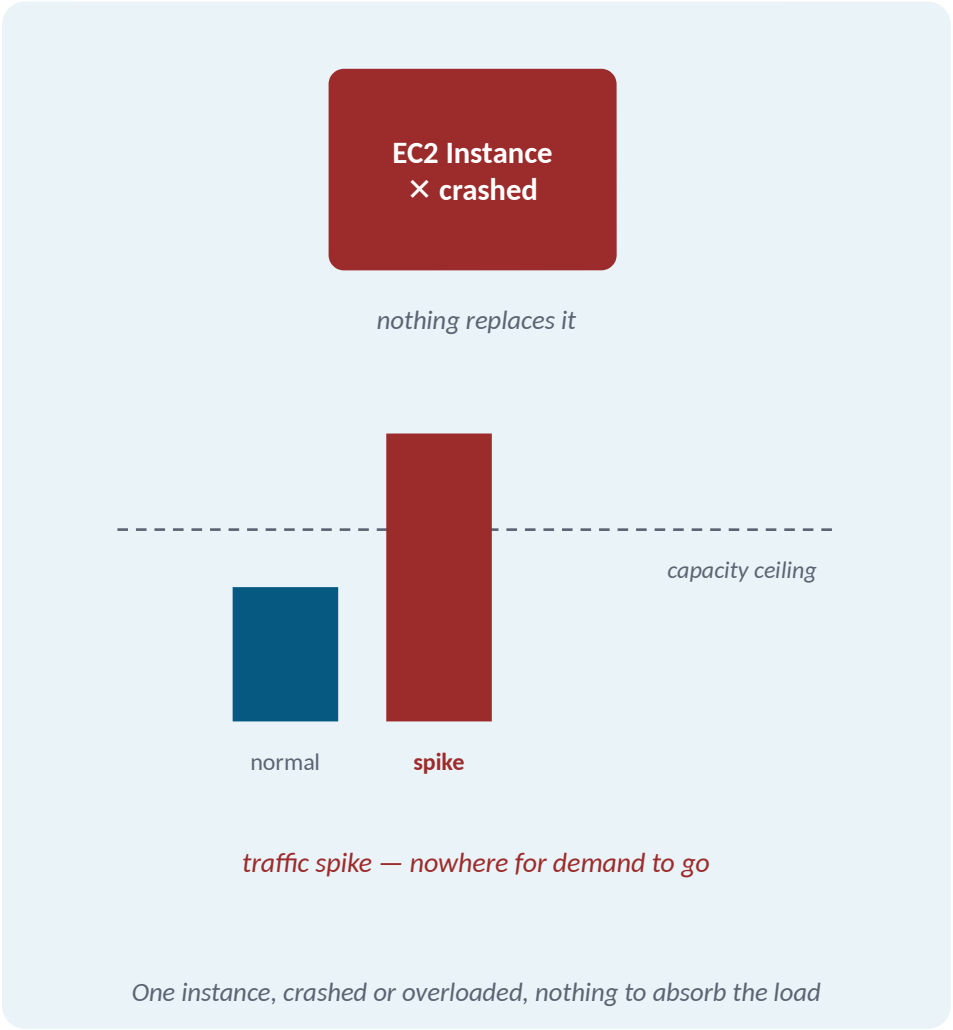
Security group filtering inbound traffic to EC2

HANDS-ON Open the port the capstone app needs

PROBLEM

What Happens When One Instance Isn't Enough?

- A single running instance is itself a single point of failure: if it crashes, nothing replaces it
- No self-healing: the application stays down until someone manually launches a new instance
- No automatic response to a traffic spike: extra demand has nowhere to go
- Scaling back down after a spike means someone watching a dashboard and terminating instances by hand

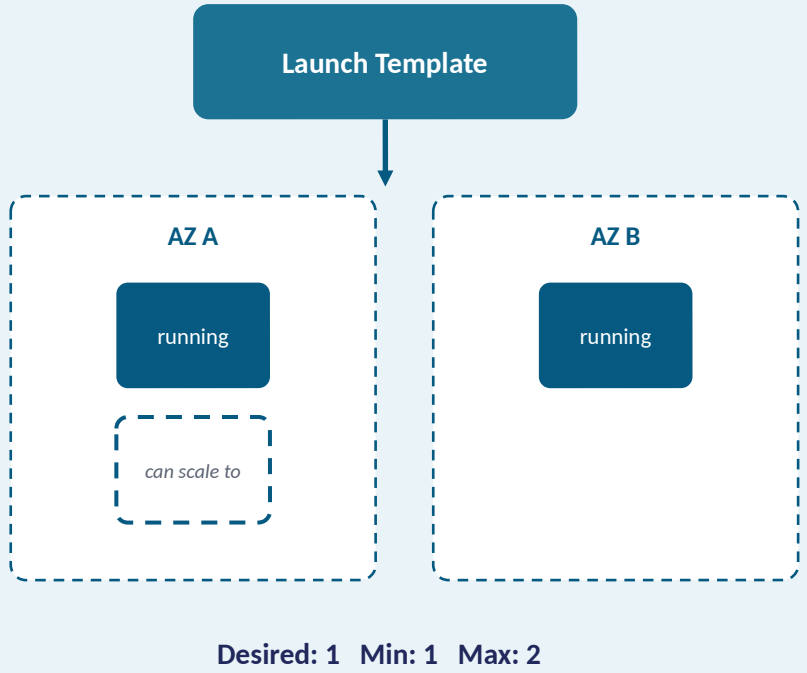


One instance, crashed or overloaded, nothing to absorb the load

SOLUTION

Auto Scaling Groups: Desired, Min, Max

- Launch template = saved, reusable instance definition (AMI, type, key pair, security groups)
- Desired capacity: how many instances should run right now
- Minimum / Maximum capacity: the floor and ceiling the group can scale between
- Spreading instances across 2+ AZs makes the fleet resilient



One launch template feeding an Auto Scaling group

HANDS-ON Configure a basic Auto Scaling group

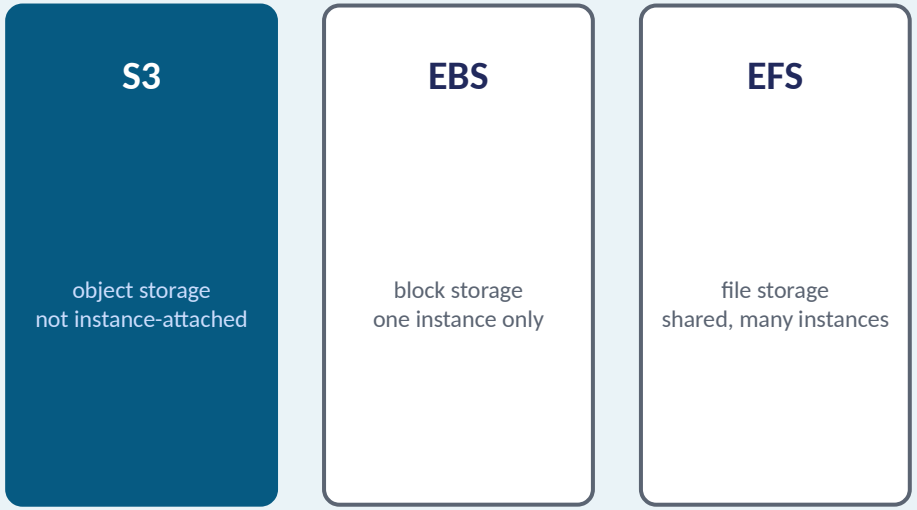
AWS Storage & Identity Essentials

WHAT'S AHEAD

- AWS's storage landscape: S3, EBS & EFS
- S3 buckets, storage classes & lifecycle policies
- IAM: users, roles, policies & permissions, least-privilege design
- Instance profiles: attaching a role to EC2
- Completing the capstone's AWS foundation

AWS's Storage Landscape

- S3 (object storage): unlimited unstructured data, accessed by API, not attached to any one instance
- EBS (block storage): behaves like a hard drive, attached to exactly one EC2 instance at a time
- EFS (file storage): a shared filesystem multiple instances can mount and use at the same time
- Today's course goes deep on one of these, S3, because it's what the capstone application needs



S3, EBS, and EFS — three different storage shapes

S3 Buckets & Storage Classes

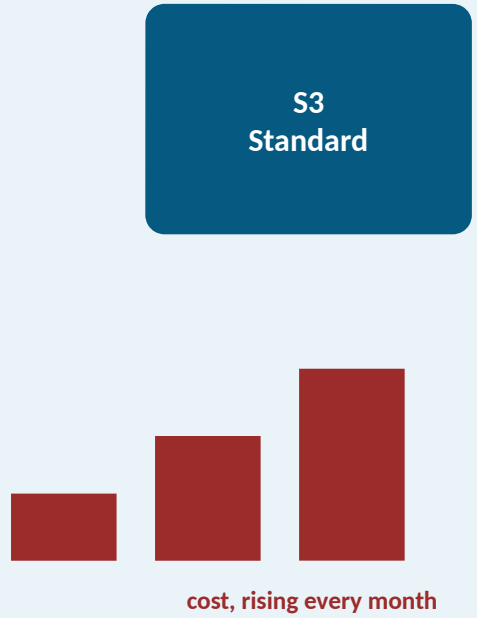
- S3 stores data as objects inside buckets — bucket names are globally unique
- A bucket's Region can't be changed after creation
- Keep the defaults: Bucket owner enforced, Block Public Access on, SSE-S3 encryption
- Storage classes trade cost vs. retrieval speed

Storage classes, fastest/costliest to cheapest

What Happens to Objects Nobody Downgrades?

PROBLEM

- Left alone, every object stays in S3 Standard, at Standard's price, forever
- Whether an object is read every second or never touched again, the bill doesn't know the difference
- Managing this by hand means someone has to remember, every month, to find and downgrade old objects
- That reliably stops happening once the person is busy, on leave, or moves to a different team



An ever-growing bucket and a rising bill nobody is watching

SOLUTION

S3 Lifecycle Policies

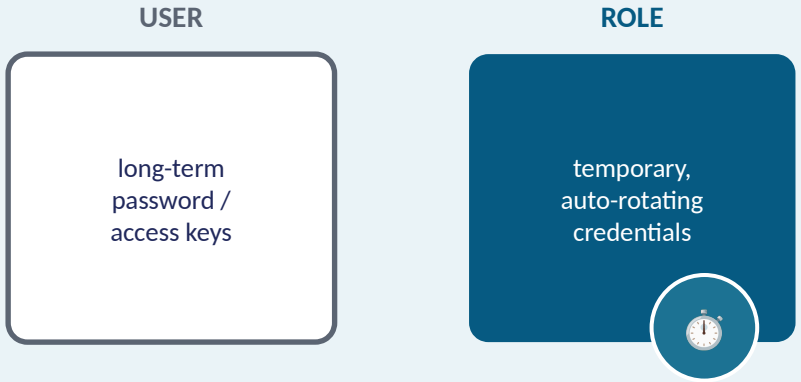
- A lifecycle rule automates moving or deleting objects over time
- Scope: all objects in the bucket, or a specific prefix/tag
- Transition action: move to a cheaper storage class after N days
- Expire action: delete the object after N days

Object lifecycle: create, transition, expire

HANDS-ON Create an S3 bucket + lifecycle policy

IAM: Users vs. Roles

- User: an identity for one person or app, with its own long-term password and/or access keys
- Role: an identity with no long-term credentials, assumed temporarily for one session
- Assuming a role issues short-term, automatically-rotating credentials, not a standing secret to leak
- AWS's guidance: workloads on AWS compute (like EC2) should use roles, not a copied user access key

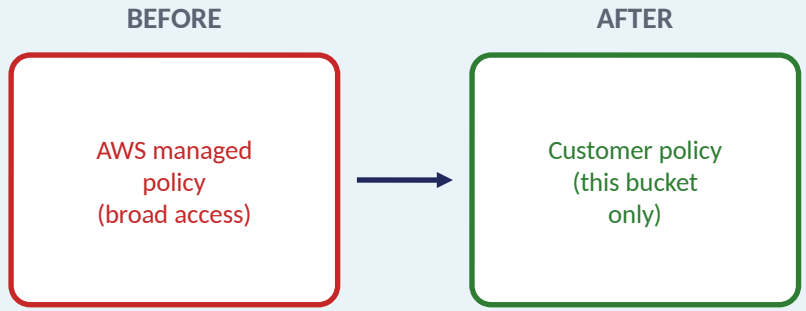


EC2 assumes the role — no access key is ever typed into the instance

A long-term user credential vs. a role's short-lived one

IAM: Policies, Permissions & Least Privilege

- Permission: one statement, this action, on this resource, allowed or denied
- Policy: a JSON document made of one or more permission statements
- A policy attaches to a user, a role, or a group, and can be reused across many identities
- Least privilege: start with a managed policy to move fast, then narrow to name only what's needed



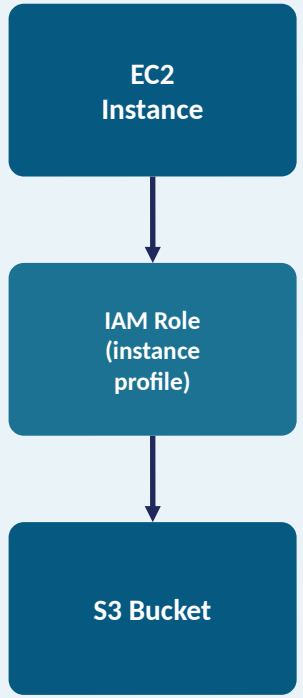
A policy = one or more permission statements. Start broad to move fast, then narrow to least privilege before the capstone depends on it.

From a broad managed policy to least privilege

HANDS-ON Create a least-privilege IAM role

Instance Profiles: Attaching a Role to EC2

- Instance profile = the container IAM builds around a role for EC2
- Only one role per instance at a time (reusable across many instances)
- Console path: Actions → Security → Modify IAM role
- Together: compute + storage + identity = the capstone's AWS foundation



The capstone's AWS foundation

EC2, IAM role, and S3 tied together

HANDS-ON Attach the role; provision the AWS foundation